

cub3D Technical Documentation

Wolfenstein-inspired raycasting engine written in C using MLX42.

Main Features

Raycasting renderer, DDA traversal, PNG texture support, flood-fill validation, collision detection, minimap rendering, portals, doors, animated enemies, sprite rendering and depth buffering.

Architecture

Parser → Validation → Engine Initialization → Game Loop → Input → Movement → Collision → Raycasting → Sprite Rendering → UI.

Build

make / make bonus